Signaling functions of code-switching in L2 learners and heritage speakers

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Code-switching in bilinguals, here defined as changing languages in the middle of a sentence, is ubiquitous. Code-switching often has a pragmatic, communicative or cognitive function. It is a way to express one's identity and helps to fill lexical gaps, but it can also signal specific types of upcoming information. Corpus and psycholinguistic studies have shown that code-switching often precedes information that is less expected (Tomić & Valdés Kroff, 2023; Myslín & Levy, 2015). Bilinguals can therefore interpret code-switching as a cue that unusual information is coming up, which facilitates the processing of this information (Tomić & Valdés Kroff, 2021).

To investigate the potential signaling function of code-switching, we conducted two cloze-completion studies with two groups of highly proficient bilinguals: (i) late bilinguals with L1 German and L2 English, and (ii) heritage speaker bilinguals with L1 Spanish and L1/L2 English. Hypotheses and analyses were preregistered at: https://osf.io/q8t7u/. Participants read highly constraining sentences with the sentence final word omitted. The material preceding the blank was either entirely in English (1a, based on Federmeier et al., 2007), or contained a switch from German to English (1b) or from Spanish to English (1c). Participants were instructed to enter the first continuation that came to mind. If code-switching indeed signals less expected content, we expected the completions to converge more strongly in the unilingual (1a) compared to the code-switching conditions. The studies were run online in PCIbex (Zehr & Schwarz, 2018) and participants were recruited via Prolific. Participants were Gen-Z and Gen-Y (Millennial) adults (18-35 years) assumed to have enhanced code-switching experience in texting. We tested 124 L1 German-L2 English speakers in Germany (upper-intermediate to high English proficiency) and 122 early Spanish-English bilinguals in the United States (Spanish heritage speakers). We also collected language background information, attitudes toward code-switching, self-reported use of code-switching, and proficiency in the non-societal language (English or Spanish).

Cloze probability analyses showed clear group specific patterns. For the German-English L2 speakers (see Figure 1), code-switching did not significantly influence the predictability of sentence completions: the cloze probability of the word completed most often did not differ between the code-switching and unilingual English conditions ($b = 0.009$, $SE = 0.01$, $p = .38$). In contrast, Spanish-English bilinguals (see Figure 2) showed lower cloze probabilities in code-switching than in unilingual English sentences ($b = 0.624$, $SE = 0.01$, $p < .0001$), in line with the predictions. In addition, the Spanish-English group produced greater lexical diversity (type-to-token ratio) in code-switching than unilingual sentences; the German-English group showed no effect of code-switching on this measure. Higher proficiency in Spanish and English was associated with a higher likelihood of producing the most predictable completion, but did not differ between code-switching and unilingual conditions. Together, the findings suggest that code-switching functions as a signaling cue for less expected information in heritage speakers, but not in late L2 learners, highlighting the potential role of bilingual experience in shaping expectation-based language processing. This will be explored further by testing Spanish heritage speakers in Germany and L1-Chinese-L2-English learners in the United States. Our ongoing research tests whether code-switching affects expectations during processing using ERP studies. If code-switching biases towards less expected information, the N400 may be smaller for the less expected condition ("collection" in 1) and larger for the expected ending ("birthday") in the code-switching than in the unilingual conditions.

(1) Experimental conditions (critical continuation omitted)

(1a)$_{\text{English}}$ He bought her a pearl necklace for her _____

(1b)$_{\text{German-English}}$ Er schenkte ihr eine Perlenkette for her _____

(1c)$_{\text{Spanish-English}}$ Le regaló un collar de perlas for her _____

(Expected: birthday; less expected alternative: collection)

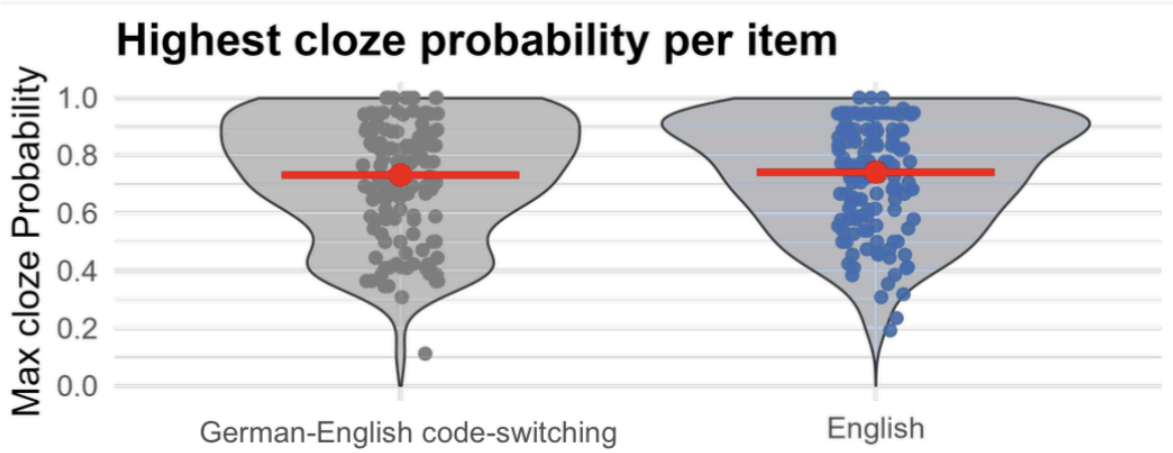

Figure 1. Highest cloze probabilities for German-English code-switching vs. English-only

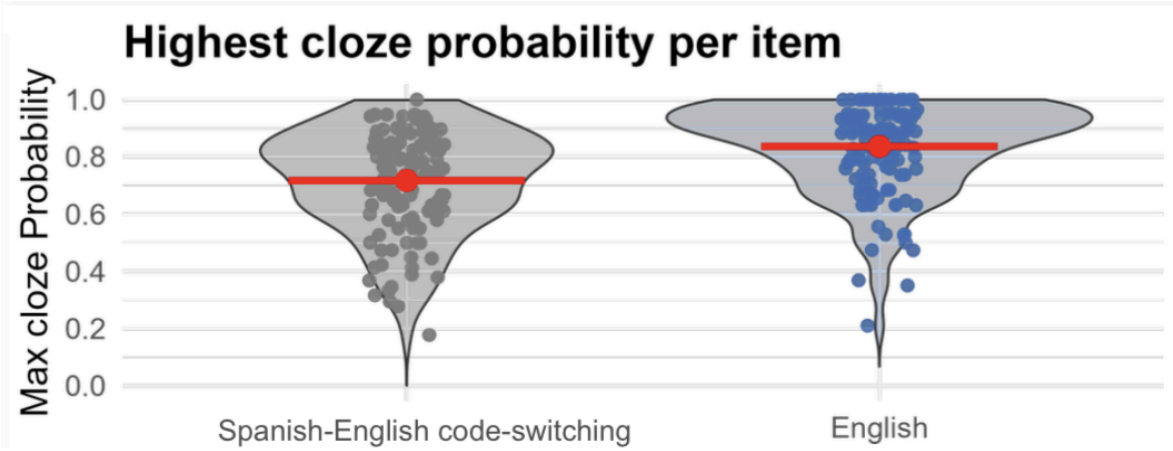

Figure 2. Highest cloze probabilities for Spanish-English code-switching vs. English-only

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